

1. Consider an array containing the following 40 integers: (10 points)

5 2 4 4 0 1 6 7 3 1 1 0 5 1 5 4 4 5 7 0 6 1 0 7 5 2 7 6 5 3 7 0 5 5 7 1 1 2 6 5

Apply counting sort to sort this array and show all the steps of your work.

2. Consider an array containing the following hexadecimal numbers: (10 points)

4EC1EEA9

520B6E78

1E90D74E

52DB6E42

5F05EF13

74284442

794E8117

55526E42

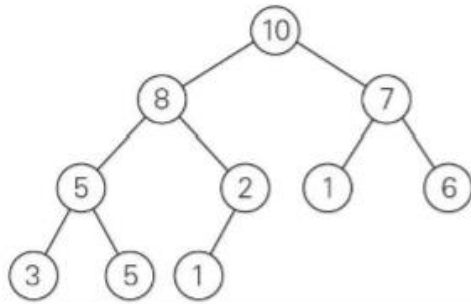
Consider you are using a version of RadixSort that sorts on one byte at a time (two hexadecimal digits at a time). Under the stated circumstances, show all the steps when you apply radix sort on this array.

3. Construct a Huffman tree (variable-length encoding) for the following: (5 points)

Symbol	A	B	C	D	E
Frequency	40	10	20	15	15

- Encode ABACABAD using the tree you generated. [1 point]
- Decode 100010111001010 using the tree you generated. [1 point]
- What compression gain (percent of improvement) do we get by using Huffman encoding (variable-length encoding) instead of a fixed-length encoding scheme. Draw the tree for the fixed-length encoding. (3 points)

4. Consider the following binary tree. Print out the nodes in the following orders (10 points):



- Traverse the tree in preorder (2 points)
 - Traverse the tree in inorder (2 points)
 - Traverse the tree in postorder (2 points)
 - How many internal nodes are there? (1 point)
 - What is the maximum width of the tree? (1 point)
 - What is the height of the tree? (1 point)
 - What is the diameter of the tree? (1 point)
5. Draw the 2-3 tree after inserting each of the following keys. **Redraw the tree for each part** (10 points)
- 50
 - 76
 - 23
 - 21
 - 20
 - 19
6. Apply Dijkstra's algorithm to compute the shortest paths starting from vertex **b** in the following graph. Show all the steps. (10 points):

